

QUANTIFYING STATE ERROR GROWTH AND CONSISTENCY OF UNCERTAINTY PROPAGATION ALGORITHMS IN CISLUNAR SPACE

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This paper aims to quantify the navigation error growth in the Cislunar space and the consistency of different uncertainty propagation algorithms in accurately capturing this error growth. The metrics for error growth include nonlinearity index and finite time Lyapunov exponent while the consistency metrics will correspond to comparing the performance of the state probability density function as approximated by different algorithms. The study includes over 17 million trajectories propagated to evaluate the behavior at a Jacobi energy level of the L_2 Lagrange point. These trajectories and corresponding uncertainty create a ground “truth”. The cislunar region from Earth to L_1 and L_2 Lagrange points is mapped in high definition. An upshot of this study also involves providing a bench-mark dataset to compare any garden variety uncertainty quantification and propagation algorithm.

Cislunar space—the region in which the primary forces on a spacecraft are the gravitational forces from the Earth and Moon—is a renewed subject of interest. The cislunar regime extends approximately twelve times farther and contains about 2000 times more volume than the geosynchronous (GEO) region. To unlock the wealth of resources the Moon holds, successful lunar missions are required. Infrastructure in cislunar space can provide surveillance capabilities in what is currently an unmonitored zone. Cislunar space is emerging as an important civil, commercial, and military domain. Nations and industry have noticed.

Cislunar space had two players during the Apollo era: the United States and Soviet Union. The Artemis era has witnessed the entry of many stakeholders. The United States, China, India, Japan, Mexico, Pakistan, Russia, South Korea, and the United Arab Emirates have operated lunar missions since 2020. Commercially lead missions have been lead by Astrobiotics, Intuitive Machines, and Lockheed Martin. NASA estimates that in the next ten years, activity in cislunar space will be equal to or exceed all that has occurred since 1957.¹ For hundreds of missions, including crewed ones, to be successful, a greater understanding of the dynamics in cislunar space is needed.

Operating in cislunar space is challenging. The environment is nonlinear, chaotic, and data sparse. One positive is that small propulsive maneuvers can greatly alter a spacecraft’s trajectory, but the

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associated propagation of navigation errors is chaotic—tiny perturbations can result in large differences. After a few days, trajectories that were almost identical can be thousands of kilometers apart. Observations in cislunar space are challenging and expensive. No singular sensor (ground station or satellite) is capable of observing the entirety of cislunar space. Most ground stations do not have the ability to observe spacecraft that far away from Earth. While many advances have been made in designing systematic ways of designing a trajectory in cislunar space^{2–4} and accurately propagating uncertainty through the underlying dynamics,^{5–7} little attention has been paid to understand the state error propagation and growth in cislunar space. A related question is: *Are there regions in cislunar space where the spacecraft motion is less sensitive to state errors, and hence neighboring trajectories remain close to each other for a long time?* The identification of such regions and understanding of departure motion dynamics will allow the design of missions for long-term surveillance in cislunar space. This paper is an attempt to answer this important research question and in this respect, the objectives of this work are twofolds:

- Understand and compute the growth of state error and chaoticity in the Circular Restricted Three Body Problem (CR3BP)
- Quantify the consistency of different uncertainty propagation algorithms in accurately capturing the growth of state errors.

PROPAGATION OF STATE ERRORS

The behavior of state error growth over time is important in Space Domain Awareness (SDA) activities, debris threat mitigation and monitoring, conjunction analysis, asteroid threat mitigation, and orbit estimation applications. Astrodynamics problems require prolonged propagation times and the accurate progression of a spacecraft’s state probability density function (PDF). Thus, due to the non-Gaussian⁸ evolution of uncertainty, linear error theory and Gaussian approximations are seldom sufficient. Error growth over time is inherently linked to the nonlinearity, stability, and chaoticity of the system. The chaoticity is highly sensitive to small changes in the initial condition.⁹ To better understand the evolution of state error growth over time, two metrics, nonlinearity index and Finite Time Lyapunov Exponent, were considered.

Nonlinearity Index

Junkins and Singla found that some coordinate system choices are more “linear” than others for evolution of the probability density function, as measured by the nonlinearity index.¹⁰ Some parts of the phase space are more linear than others. Nonlinearity index measures the departure of trajectories and can be used to characterize the linearity and chaoticity of a dynamical system. To explain nonlinearity index, consider the following system describing the motion of a spacecraft in the cislunar regime,

$$\dot{\mathbf{x}} = f(\mathbf{x}) \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^6$ represents the position and velocity of the spacecraft. Let $\bar{\mathbf{x}}(t)$ denote the spacecraft trajectory that results from solving Eq. (1) with the nominal initial condition, $\bar{\mathbf{x}}(t_0)$. The worst-case distribution of initial conditions centered around the nominal initial condition, $\bar{\mathbf{x}}(t_0)$, are considered. Generally, the set of worst-case $\delta\mathbf{x}(t_0)$ -variations are defined to be the set of points lying on the surface of a $k-\sigma$ ellipsoid associated with the uncertainty of the $\delta\mathbf{x}(t_0)$ -variations. These variations,

however, are restricted to being near-uniformly spaced on a worst-case surface. In this study, a hyper sphere with $\|\delta \mathbf{x}_i(t_0)\| = \delta \mathbf{x}_{\max}$. More generally, the worst-case surface could be an ellipsoid, but an ellipsoid can be mapped into a sphere via coordinate rotations and scaling.

The deviation of the i^{th} trajectory from the nominal trajectory is given by,

$$\delta \bar{\mathbf{x}}_i(t) = \mathbf{x}_i(t) - \bar{\mathbf{x}}(t) \approx \bar{\Phi}(t, t_0) \delta \bar{\mathbf{x}}_i(t_0) \quad (2)$$

where $\bar{\Phi}(t, t_0)$ is the state transition matrix computed along the nominal trajectory, $\bar{\mathbf{x}}(t)$. Similarly, the deviation of the nominal from the i^{th} trajectory can be written as,

$$\delta \mathbf{x}_i(t) = \bar{\mathbf{x}}(t) - \mathbf{x}_i(t) \approx \Phi_i(t, t_0) \delta \mathbf{x}_i(t_0) \quad (3)$$

where $\Phi_i(t, t_0)$ is the state transition matrix computed along the i^{th} trajectory, $\mathbf{x}_i(t)$. Note that the first order approximations of $\delta \bar{\mathbf{x}}_i(t)$ and $\delta \mathbf{x}_i(t)$ will be the same for the linear system, but they will differ for a nonlinear system. Hence, one can define the measure of nonlinearity to be,¹⁰

$$\nu(t, t_0) \triangleq \sup_{i=1 \dots N} \frac{\|(\Phi_i(t, t_0) - \bar{\Phi}(t, t_0)) \delta \mathbf{x}_i(t_0)\|}{\|\bar{\Phi}(t, t_0) \delta \mathbf{x}_i(t_0)\|} \quad (4)$$

$$= \sup_{i=1 \dots N} \frac{\|\Phi_i(t, t_0) - \bar{\Phi}(t, t_0)\|}{\|\bar{\Phi}(t, t_0)\|} \quad (5)$$

Notice that the nonlinearity index defined in Eq. (5) is itself a non-dimensional measure of the variation of the state-transition matrices over a set of N extreme points. Note that a $\nu(t, t_0)$ of order 10^{-2} or smaller qualitatively indicates “less than 1%” nonlinearity of the Eq. (1) while a $\nu(t, t_0)$ of order 1 qualitatively indicates a “100%” worst-case departure from the linear approximation when mapping worst-case variations in the vicinity of $\bar{\mathbf{x}}(t_0)$. The nonlinearity index of Eq. (5) is a deterministic measure associated with a particular domain of initial conditions and a particular interval of time.

Finite Time Lyapunov Exponent

The Lyapunov exponent is an important metric in the study of dynamical systems. It is defined as the average rate of divergence (convergence) of two neighboring trajectories.

$$\lambda_k = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|\delta \mathbf{x}_{i_k}(t)|}{|\delta \mathbf{x}_{i_k}(t_0)|} = \lim_{t \rightarrow \infty} \frac{1}{t} \ln |\bar{\Phi}_k(t, t_0)|, k = 1 \dots n \quad (6)$$

Where, $\bar{\Phi}_k$ denotes the k^{th} row of state transition matrix, $\bar{\Phi}$. The number of Lyapunov exponents is equal to the number of dimensions of the phase space (n), but it is common to just refer to the largest one. Therefore, if Λ is defined to be the maximum of all possible Lyapunov exponents, then Λ can be written as:

$$\Lambda = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \|\bar{\Phi}\|_2 \quad (7)$$

Therefore, the nonlinearity index, $\nu(t, t_0)$, can be related to Λ as:

$$\lim_{t \rightarrow \infty} \nu(t, t_0) \sim \sup_{i=1 \dots N} \frac{e^{\Lambda_i} - e^{\bar{\Lambda}}}{e^{\bar{\Lambda}}} \quad (8)$$

Where Λ_i is evaluated along the i^{th} trajectory ($\mathbf{x}_i$). The main drawback of the Lyapunov exponent is that it is defined for infinite time. Thus it does not reflect the rate of divergence (convergence) for short time period whereas nonlinearity index defined in equation (5) can be used to measure rate of divergence (convergence) for different parts of a trajectory.

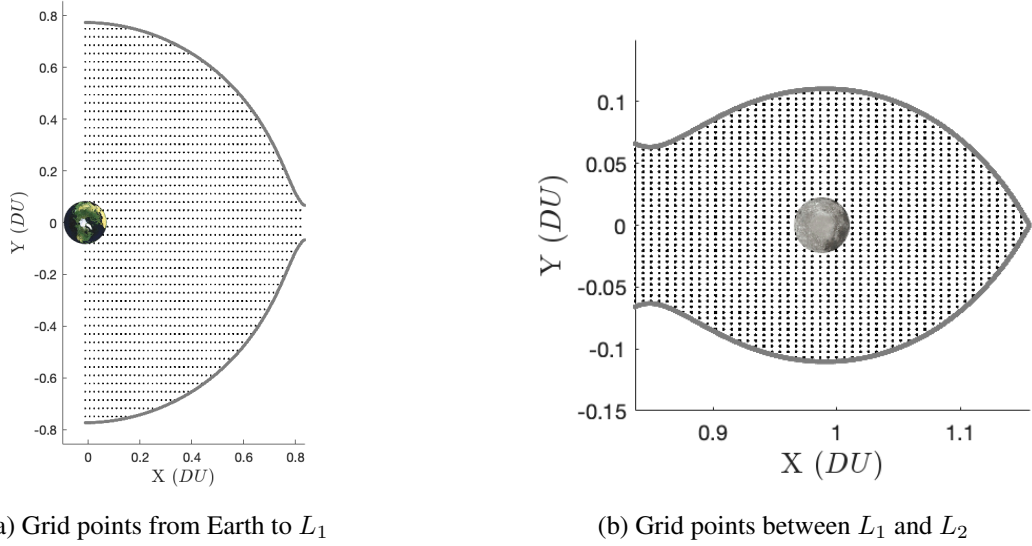


Figure 2: Grid points bounded by Jacobi energy of L_2

ity index calculation. The worst-case perturbed points were defined as

$$x_p(t_0) = x_N(t_0) + \delta x, y_p(t_0) = y_N(t_0) + \delta y, \mathbf{v}_p = \mathbf{v}_N + \delta \mathbf{v} \quad (10)$$

The maximum position perturbation (δx and δy) was set to 5km. The angle between the nominal velocity, $\mathbf{v}_N$, and $\delta \mathbf{v}$, was determined using θ —a error term in a third dimension. θ was mapped to lie between $-\pi$ and π . To generate the perturbed initial conditions, 5,000 uniformly distributed points on an hyper sphere in δx - δy - θ space were generated according to the algorithm described in [10]. Note, the 5000 perturbed points all lie on the edge of the uncertainty bubble. Thus, they represent worst case scenarios. It does not produce a probability density function like a Monte Carlo simulation. Equation 9 was used to compute the magnitude of the perturbed velocity vector, v_p , given the perturbed position (x_p, y_p) . The following equation was used to compute the magnitude of δv for each perturbed initial condition:

$$(\delta v)^2 + 2\delta v (v_{x_N} \cos \theta + v_{y_N} \sin \theta) + v_N^2 - v_P^2 = 0, \quad (11)$$

where v_{x_N} and v_{y_N} are the nominal x and y velocities. There are two solutions to Eq (11). The solution with the smaller magnitude was chosen.

This velocity formulation occasionally resulted in an imaginary solution for $\delta \mathbf{v}$. This occurred when

$$(v_{x_N} \cos(\theta) + v_{y_N} \sin(\theta))^2 - v_N^2 + v_P^2 < 0 \quad (12)$$

This occurred when the perturbed velocity, v_P , was smaller than the nominal velocity, v_N , and $\theta \approx \pi/2, -\pi/2$. When this occurred, the Jacobi energy constraint on the perturbed velocity was relaxed so that Eq. (12) would equal zero. This dilemma did not occur often. The difference between the desired Jacobi energy (that of L_2) and actual Jacobi energy was calculated for each perturbed trajectory. The mean error for over all grid points was 6.73×10^{-7} .

The nominal and perturbed states along with their state transition matrices were numerically integrated for 30 days using the CR3BP dynamical model. The state and state transition matrix

were recorded every 30 minutes. MATLAB's *ode113* was used to propagate with absolute and relative tolerances of 10^{-10} .

This analysis required the propagation of **17,913,582 trajectories** to compute the nonlinearity index for each grid point. The FTLE calculation only requires the propagation of the 3582 nominal trajectories. The FTLE along a nominal trajectory reflects the first-order approximation of the rate of growth of error. The nonlinearity index at a given time measures the worst-case variations in the state transition matrix (the sensitivity of the state to initial condition) and hence, the variation in the growth of error when mapping worst-case variations in the vicinity of the nominal initial condition.

The nominal states were initialized to be at a periaipse condition. There were still trajectories that passed through the Earth or Moon. Trajectories were allowed to do so. There were also trajectories, however, that came so close to the point-mass Earth or point-mass Moon that the integration failed. Only one nominal trajectory suffered this. The integration of perturbed trajectories failed far more often. If a significant quantity of a grid point's perturbed trajectories have failed, the nonlinearity index calculation is unreliable. Thus, the grid point was considered "failed" if 50 or more of the perturbed trajectories failed to integrate. The nonlinearity index after the time of failure was set to the last value before failure. This did not occur at high frequency. Of the 3582 grid points, only 29 failed.

REPRESENTATION OF NONLINEARITY AND FTLE IN CISLUNAR SPACE

The nonlinearity index and FTLE were computed every thirty minutes along a nominal trajectory corresponding to a specific grid point. The nonlinearity index and FTLE values were assigned to the grid point the nominal trajectory started at. Additionally, the root-mean-square (RMS) of nonlinearity index was computed for visualization purposes. The RMS of a quantity $m(t)$ is computed

as: $RMS(m) = \sqrt{\frac{1}{N} \sum_{i=1}^N m^2(t_i)}$. The RMS of nonlinearity index represents the nonlinearity over a time period instead of a snapshot in time. Figures (3) and (5) show contour plots of the nonlinearity index, RMS of nonlinearity index, and FTLE at 15 and 30 days in the Earth to L_1 region and L_1 to L_2 region. The nonlinearity index results were more consistent over the region. Therefore, nonlinearity index was the primary metric considered for analysis.

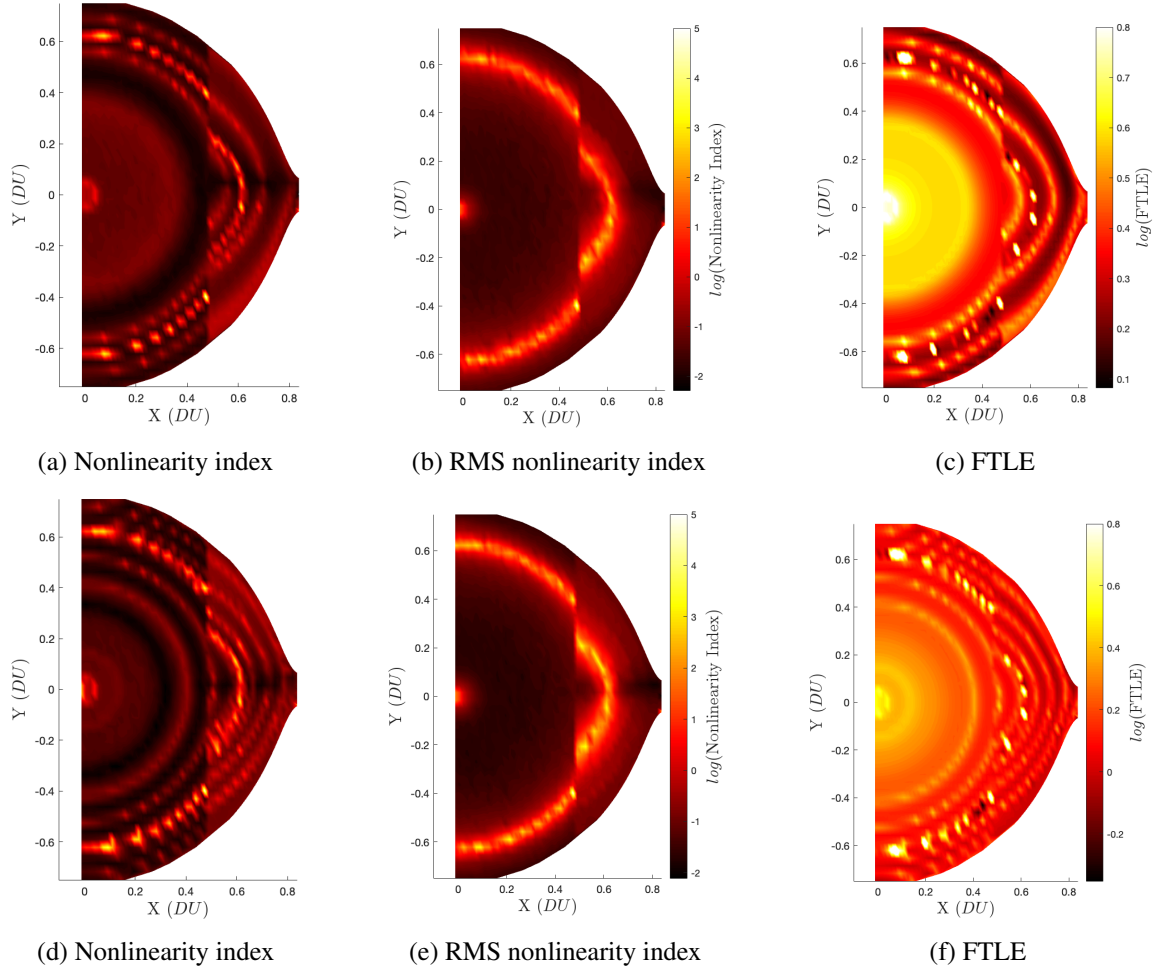
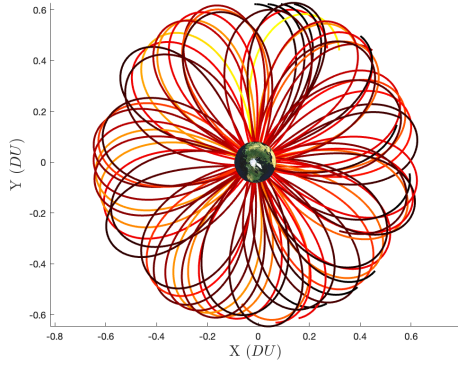


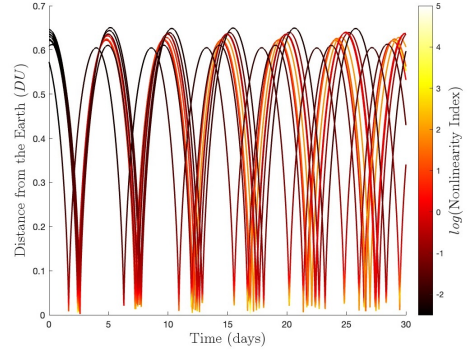
Figure 3: Heat map from Earth to L_1 at 15 and 30 days. 3a 3b 3c are at 15 days. 3d 3e 3f are at 30 days.

There is one notable structure in the Earth region. Best shown in Figure (3d), there is a ring of high nonlinearity index centered at the Earth with an approximate radius of 0.6 DU . The ring does have a discontinuity. The discontinuity occurs along the line where the closer primary body flips between the Earth and the Moon. This is not surprising as the closer primary body determines the velocity direction. A select group of trajectories from the ring is shown in Figure (4a).

The trajectories are colored by the logarithm of their instantaneous nonlinearity index. The shape of the trajectories is consistent. These trajectories perform many passes by (or inside) the Earth. This trend is present throughout the grid points. Many close approaches to a primary body is correlated with high nonlinearity index. Figure (4b) displays the distance from the Earth over time for the ring trajectories. The coloring shows that trajectories consistently have the highest nonlinearity index in close proximity to the Earth.



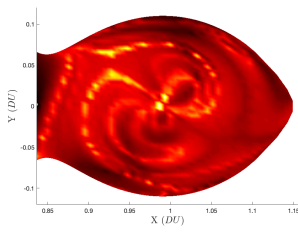
(a) Ring trajectories



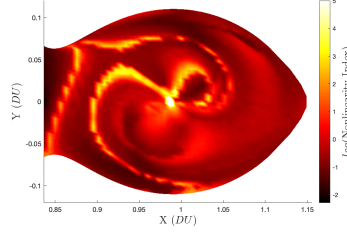
(b) Distance from the Earth

Figure 4: Ring of high nonlinearity index

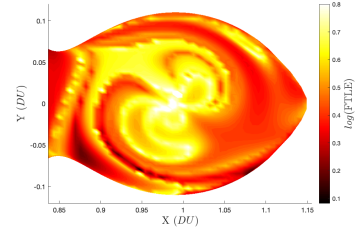
The heat map in the L_1 to L_2 region is far more interesting. There is a spiral of high nonlinearity index areas. There are high and low nonlinearity region in close proximity. There is a diagonal line of high nonlinearity separating the region from the L_1 gateway. The highest nonlinearity area is right at the Moon, but interestingly, there exists low nonlinearity regions by the Moon almost entirely engulfed by high nonlinearity regions. The analysis of the L_1 to L_2 is not as simple as the Earth region.



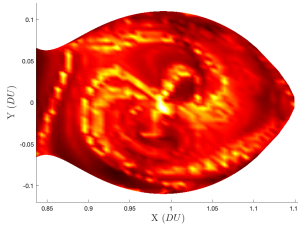
(a) Nonlinearity index



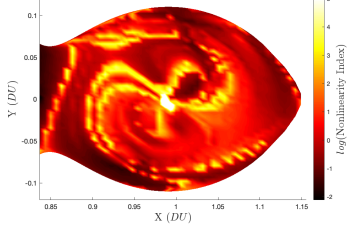
(b) RMS nonlinearity index



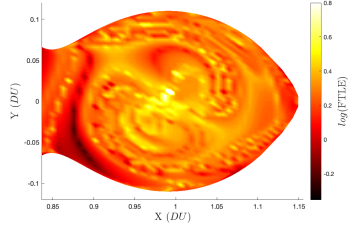
(c) FTLE



(d) Nonlinearity index



(e) RMS nonlinearity index



(f) FTLE

Figure 5: Heat map from L_1 to L_2 at 15 and 30 days. 5a 5b 5c are at 15 days. 5d 5e 5f are at 30 days.

The study of the nonlinearity index and the FTLE for cislunar trajectories provide the following observations:

For Earth Neighborhood:

1. The nonlinearity index, derived from the divergence of nominal and perturbed trajectories,

shows a ring of high values centered on Earth with a radius of approximately 0.6 DU.

2. High nonlinearity is strongly associated with trajectories that make multiple close approaches to Earth. Such encounters amplify nonlinear effects as perturbations grow significantly over time.
3. The FTLE heat map mirrors the nonlinearity index trends, highlighting regions where small differences in initial conditions lead to exponential divergence of trajectories. High FTLE values are concentrated near Earth, particularly where close approaches occur, indicating locally chaotic behavior.
4. The RMS nonlinearity index, which averages nonlinearity over time, provides a broader temporal perspective of dynamical sensitivity. It emphasizes persistent regions of high nonlinearity, aligning with the instantaneous patterns but smoothing transient features.

For Moon neighborhood:

1. A spiral structure of alternating high and low nonlinearity regions is observed, showcasing the complexity of this region's dynamics.
2. A distinct diagonal band of high nonlinearity separates the L_1 gateway region, signifying zones of enhanced sensitivity where small perturbations can significantly alter trajectories.
3. Near the Moon, the highest nonlinearity values are found, consistent with strong gravitational interactions. However, isolated pockets of low nonlinearity are embedded within these high nonlinearity regions, suggesting locally stable zones in an otherwise chaotic environment.
4. The FTLE map reveals a similar spiral structure, supporting the observation that this region exhibits intricate sensitivity to initial conditions. High FTLE values near the Moon align with the gravitational dominance of the Moon and its impact on local trajectories.

The Moon neighborhood is particularly interesting because of the rich dynamical behavior exhibited through the analysis of nonlinearity and the FTLE. This leads one to hypothesize that there are certain groups of dynamical behaviors associated with their nonlinearity indices. The following section investigates the particular geometry of trajectories originating in the lunar vicinity bounded by the L_1 gateway.

Grouping by Geometry

There are 1810 grid points inside the L_1 to L_2 region. Only 746 of these grid points had trajectories that ended in the L_1 to L_2 region. Approximately 60% of the trajectories transit through the L_1 gateway. There are no trajectories that started in the Earth to L_1 region that entered the L_1 to L_2 region. The nonlinearity index of grid points inside the L_1 to L_2 region is higher than the Earth to L_1 region. Of the 600 grid points with the lowest RMS of nonlinearity index, only 41 were located in the L_1 to L_2 region.

The **1810 trajectories** were manually classified into **nine groups**. The first five Groups exited the L_1 to L_2 region through the L_1 gateway. Classification is based on the **number of close approaches or orbits of the Moon** before exiting as follows:

- **Group 0:** No orbits around the Moon.
- **Group 1:** One orbit around the Moon.
- **Group 2:** Two orbits around the Moon.
- **Group 3:** Three orbits around the Moon.
- **Group 4:** Four or more orbits around the Moon.

Last Four Groups (Remaining Inside L_1 to L_2 Region): Trajectories in groups five through 8 stayed confined within the L_1 to L_2 region. Classification is based on the following characteristics:

- **Group 5: Periodic:** Trajectories that repeated in a periodic manner.
- **Group 6: Almost Periodic:** Trajectories that were close to periodic but not exact.
- **Group 7: Approaching the Zero Velocity Curve:** Trajectories with sharp turns due to approaching the zero velocity curve and reversing direction.
- **Group 8: Other:** Trajectories that did not fit into the above categories.

The groups are shown in Figure (6).

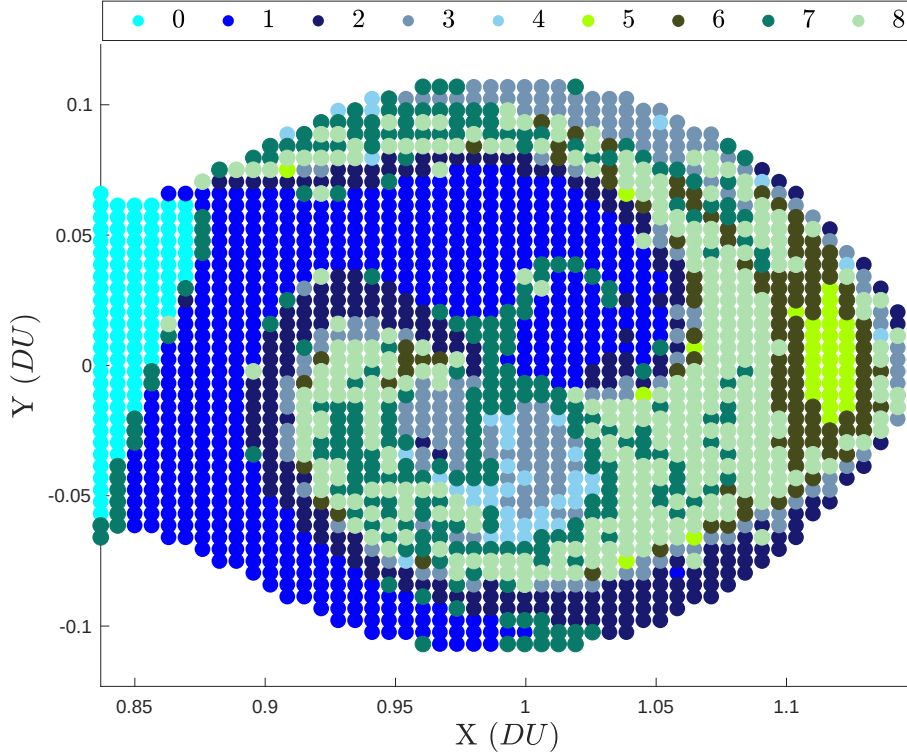
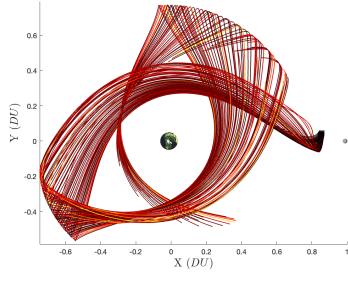
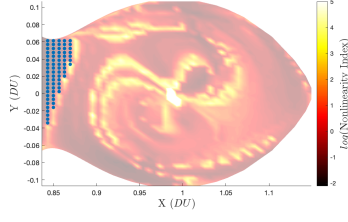


Figure 6: L_1 to L_2 grid points by group. Groups 0 to 4 left the L_1 to L_2 region and are number by orbits of the Moon prior to leaving. Groups 5 to 8 stay within the L_1 to L_2 region. The groups are periodic, almost periodic, zero velocity curve approaching, and other.

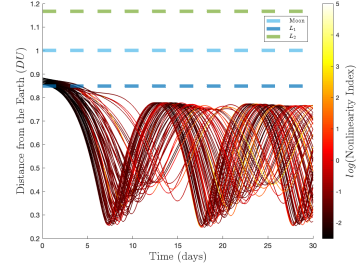
Figure (7) shows the trajectories, color coded by nonlinearity index, starting grid points overlapped on the nonlinearity index heat map, and distance from Earth over time for groups 0 to 4.



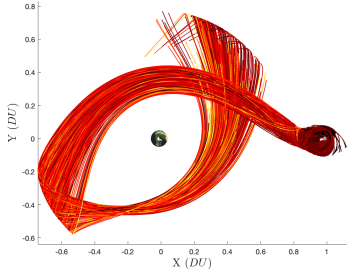
(a) Group 0 trajectory



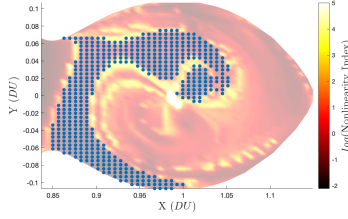
(b) Group 0 grid points



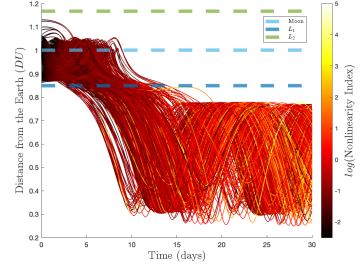
(c) Group 0 distance from Earth



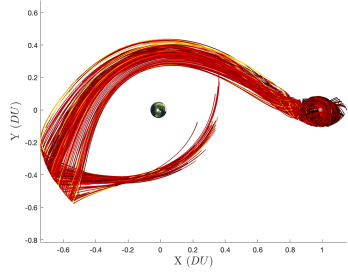
(d) Group 1 trajectory



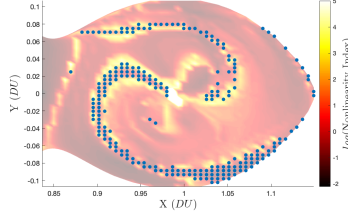
(e) Group 1 grid points



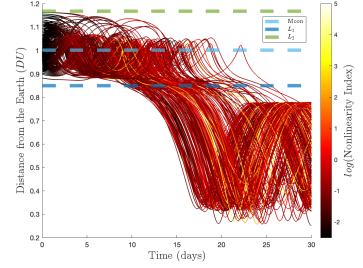
(f) Group 1 distance from Earth



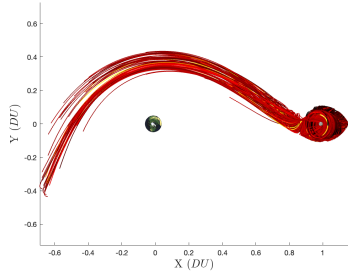
(g) Group 2 trajectory



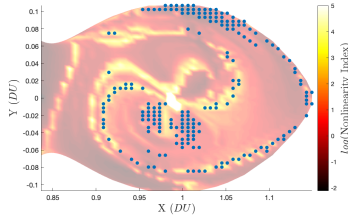
(h) Group 2 grid points



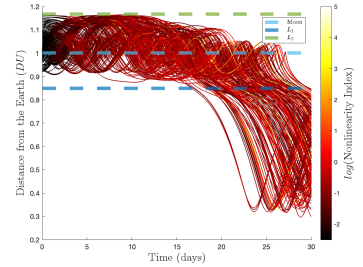
(i) Group 2 distance from Earth



(j) Group 3 trajectory



(k) Group 3 grid points



(l) Group 3 distance from Earth

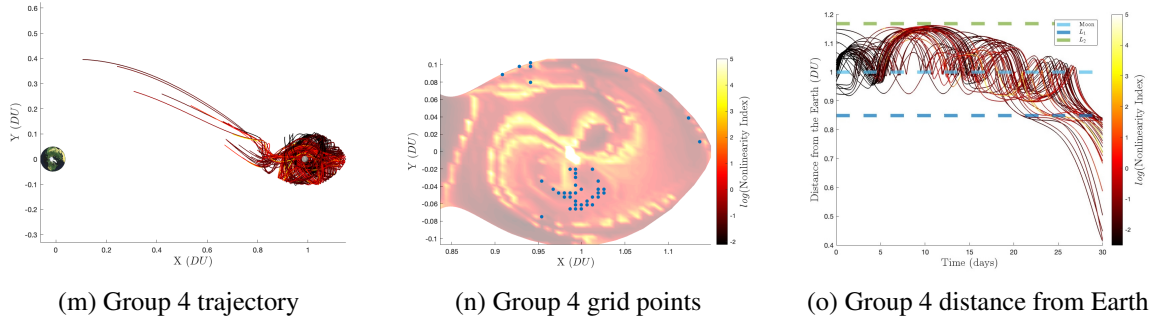
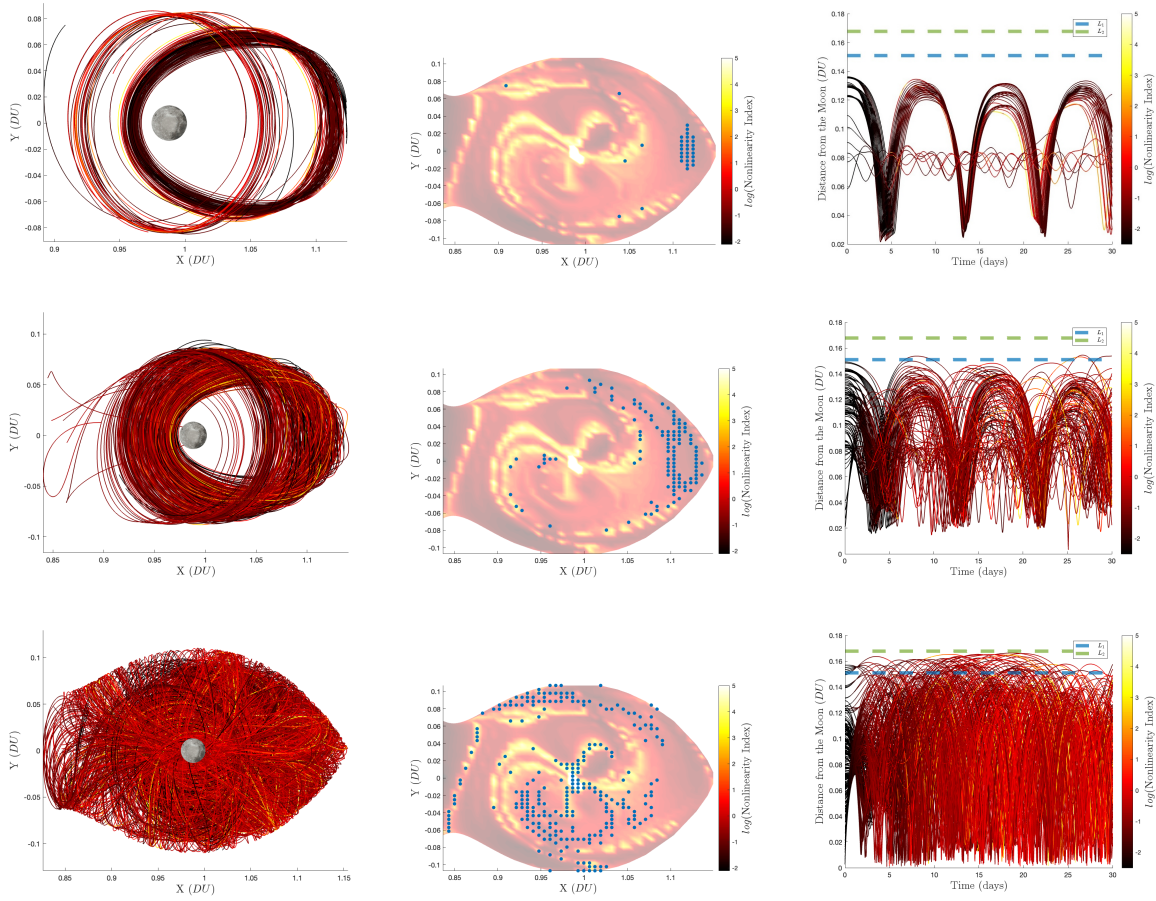


Figure 7: Trajectory, grid points, and distance from Earth for groups 0-4. All groups leave L_1 to L_2 region

Figure (8) shows the trajectories, color coded by nonlinearity index, starting grid points overlapped on the nonlinearity index heat map, and distance from Earth over time for groups 5 to 8.



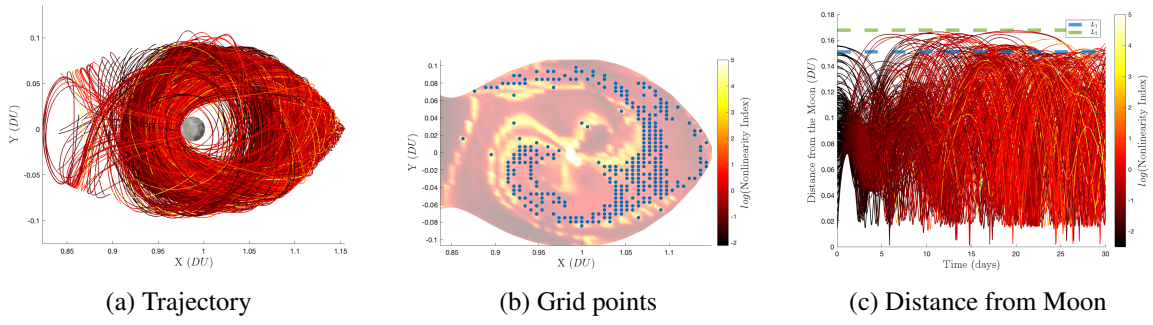


Figure 8: Nonlinearity index heat map with open L_2

The nonlinearity index of the group was analyzed and are tabulated below.

Group	0	1	2	3	4	5	6	7	8
Mean	-0.91	-0.02	0.8810	0.786	0.49	-0.02	0.058	1.49	0.83
Min	-2.11	-1.59	-1.08	-1.26	-0.88	-0.76	-1.08	-0.84	-1.30
Max	5.5192	4.84	5.16	4.54	3.36	3.43	3.91	13.13	4.54

Table 1: Mean logarithm of RMS of nonlinearity index up to 30 days

It is evident from the figures above that the nonlinearity index provides a way to cluster trajectories based on their geometry. One can make the following observations based on the figures above:

1. Trajectories that spend more time in the L_1 to L_2 region have higher nonlinearity. Therefore, those trajectories that remain bounded to the moon neighborhood are often those that accumulate nonlinearity over time. This is primarily due to the multiple close passes of these trajectories to the Moon.
2. Close approaches to the Earth or Moon increase nonlinearity. In the Moon region, the group with the highest nonlinearity was the group that approach the zero velocity curves. Trajectories in this group were, similar to trajectories in the ring and Earth, flying by the Moon many times.
3. The nearly periodic trajectory group lies inside the larger quasi periodic group. This map was generated for a single Jacobi energy. Therefore, there is a high possibility of a perfectly periodic orbit residing at the center of this cluster. This is shown in Figure (6).
4. The structure of the L_1 gateway is visible through a strong barrier of large nonlinearity and FTLE. It may also be postulated that this barrier is indeed the gateway itself. Trajectories immediately to the left of this barrier exit the gateway without entering the moon neighborhood.
5. Trajectories that are "weakly" bound to the lunar gravity are those that lie in low nonlinearity regions. They depart the gateway without experiencing many close approaches to the Moon.
6. As trajectories spend more time in the Moon region, they accumulate nonlinearity.

Having discussed the different geometries of trajectories that can arise in the cislunar region, it is pertinent to investigate the growth of uncertainty in these regions. Well-known algorithms are compared with recently developed non-product quadrature rules for state uncertainty propagation to study the consistency of such algorithms. The following sections discuss these metrics for different trajectory geometries.

UNCERTAINTY PROPAGATION ALGORITHMS

The second part of the study tests multiple uncertainty propagation algorithms. The uncertainty propagation algorithms seek to replicate the performance of the 5000 perturbed sphere points, but doing so with less computational time. Additionally, the 5000 sphere points do not produce a probability density function. The uncertainty propagation algorithms do. Five uncertainty propagation algorithms were used: linear propagation, unscented transform (UT), Conjugate Unscented Transform (CUT) 4, CUT 6, and CUT 8.

Linear Propagation

The linear propagation method uses a Taylor series expansion to the d^{th} order with $\mathbf{x}$ as the input variable and $\mathbf{y}$ as the output variable, the following equation from is applied:¹¹

$$\mathbf{y}^{(d)} = f^{(d)}(\mathbf{x}^* + \delta\mathbf{x}) \approx \underbrace{f(\mathbf{x}^*)}_{\text{Nominal Solution}} + \underbrace{\frac{\partial f(\mathbf{x}^*)}{\partial x_{q_1}}}_{1^{st} \text{ Order Sensitivity}} + \underbrace{\frac{1}{2!} \frac{\partial^2 f(\mathbf{x}^*)}{\partial x_{q_1} \partial x_{q_2}} \delta x_{q_1} \delta x_{q_2} \dots}_{2^{nd} \text{ Order Sensitivity}} + \underbrace{\frac{1}{d!} \frac{\partial^d f(\mathbf{x}^*)}{\partial x_{q_1} \partial x_{q_2} \dots \partial x_{q_d}}}_{\text{Higher Order Terms}}, \text{ for } q_1, q_2, \dots, q_d = 1, 2 \dots n \quad (13)$$

If the expression for $\mathbf{y}$ is simplified to the first order, the mean and covariance are the following:

$$\mu_{\mathbf{x}} = E[y^{(1)}] = E[f(\mathbf{x}^*)] + \mathbf{H}E[\delta\mathbf{x}] = f(\mathbf{x}^*), \mathbf{H} = \frac{\partial f(\mathbf{x}^*)}{\partial \mathbf{x}} \quad (14)$$

$$\Sigma_{\mathbf{y}} = E\left[\left(y^{(1)} - \mu_{\mathbf{y}}\right)\left(y^{(1)} - \mu_{\mathbf{y}}\right)^T\right] = E[\mathbf{H}\delta\mathbf{x}\delta\mathbf{x}^T\mathbf{H}^T] = \mathbf{H}\Sigma_{\mathbf{x}}\mathbf{H}^T \quad (15)$$

The generalized Eq. (14) and Eq. (15) is the linear propagation method. The mean and covariance are simultaneously propagated. The covariance is propagated with the following equation:

$$\dot{\Sigma} = \mathbf{A}\Sigma + \Sigma\mathbf{A}^T \quad (16)$$

where $\mathbf{A}$ is the monodromy matrix.

Unscented Transform

The unscented transform (UT) builds upon the linear propagation method. The UT is similarly formulated, however, with higher order. UT algorithm used multiple points to try and capture the dynamics. Singla and Junkins outlined the method to calculate the initial points. The mean and covariance are calculated with the following expectation integrals:¹⁰

$$\mu = E[x] = \int_{-\infty}^{+\infty} xp(x)dx \approx \frac{1}{N} \sum_{i=1}^N x_i \quad (17)$$

$$P = E[(x - \mu)(x - \mu)^T] = E[xx^T] - \mu\mu^T \approx \frac{1}{N} \sum_{i=1}^N (x_i - \mu)(x_i - \mu)^T \quad (18)$$

Where $x^{(k)}$ and $\mu^{(k)}$ are the k^{th} components of the i^{th} sample and the sample mean, respectively. For a d dimensional system, the UT generates $N_s = 2d + 1$ Sigma Points. The Sigma Points can be generated at the initial time as follows:

$$\chi_1 = \mu_0 \quad (19)$$

$$\chi_i = \mu_0 + \left[\sqrt{(d + \kappa)P_0} \right]_i, \quad i = 2, 3, \dots, d + 1 \quad (20)$$

$$\chi_i = \mu_0 - \left[\sqrt{(d + \kappa)P_0} \right]_i, \quad i = d + 2, d + 3, \dots, 2d + 1 \quad (21)$$

Where μ_0 , P_0 are the initial mean and covariance, respectively, κ is a parameter controlling the spread of the Sigma Points, and $\left[\sqrt{(d + \kappa)P_0} \right]_i$ is the i^{th} column of $\sqrt{(d + \kappa)P_0}$. The UT Sigma Points are also assigned the following weights: $W_1 = \frac{\kappa}{d + \kappa}$, $W_i = \frac{1}{2(d + \kappa)}$, $i = 2, 3, \dots, 2d + 1$

For the simulations carried out in this paper, $\kappa = 0$. Finally, the mean and covariance of the UT Sigma Points can be computed as follows:

$$\mu_{UT} = \sum_{i=1}^{N_s} W_i \chi_i \quad (22)$$

$$P_{UT} = \sum_{i=1}^{N_s} W_i (\chi_i - \mu)(\chi_i - \mu)^T \quad (23)$$

The initial covariance, P_0 , was calculated using the 5000 perturbed sphere points. The total range of perturbed points in each dimension was set to be 6σ . The initial covariance was then calculated using σ . This initial covariance was used for the UT, and CUT methods.

Conjugate Unscented Transform

The Conjugate Unscented Transform (CUT) method further builds upon UT. This method exploits the structure of the domain to choose specially defined axes on which quadrature points are selected. These CUT points are developed using minimal cubature rules and offer similar orders of accuracy as the Gaussian quadrature method but with fewer required points. In this method, the points are chosen symmetrically at conjugate axes in addition to the principal axes, i.e. the UT axes. The initial points are generated with a quadrature method.¹¹ The CUT method of selecting states and weights is derived from replacing $f(\mathbf{x}_i)$ in Eq. (24) with a Taylor series approximation as shown in Eq. (13).

$$\mu = E[f(\mathbf{x})] = \sum_{i=1}^N w_i f(\mathbf{x}_i) \quad (24)$$

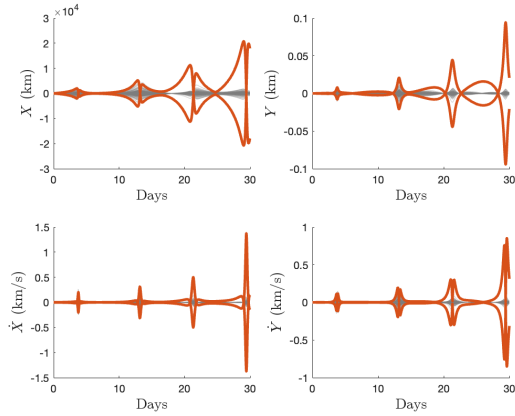
The mean and covariance was calculated at each time period in the same fashion as the UT. Three different orders of CUT were used: CUT 4, CUT 6, and CUT 8. CUT is the most computation of the uncertainty propagation methods. Specifically, CUT 8 is the most computation. CUT 8, however, has the most points. Thus, it is expected to perform the best.

Propagation Methods applied to select grid points

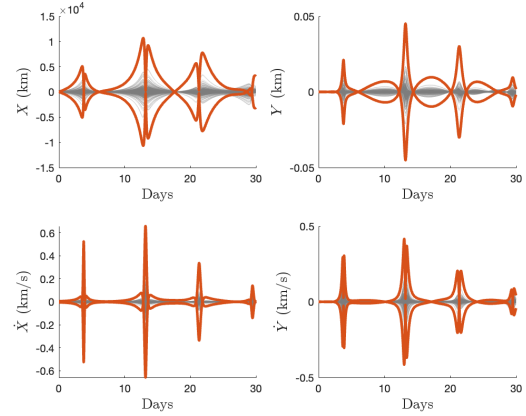
The uncertainty propagation algorithms were tested on select grid points of interest. The first point tested was one from the periodic group of grid points. The logarithm of the RMS of the nonlinearity index up to 30 days of the trajectory is -0.7562. This is a very low nonlinearity. The mean and covariance of each uncertainty propagation method was calculated. Figure (9) shows the distance from the mean of the 5000 sphere points as well as the 3σ bounds based on the covariance.

The second grid point tested was one with a medium nonlinearity index. A grid point that performed one orbit around the Moon before leaving to the Earth region was tested. The logarithm of the RMS of the nonlinearity index up to 30 days of the trajectory is 1.77. Figure (10) shows the distance from the mean of the 5000 sphere points as well as the 3σ bounds based on the covariance.

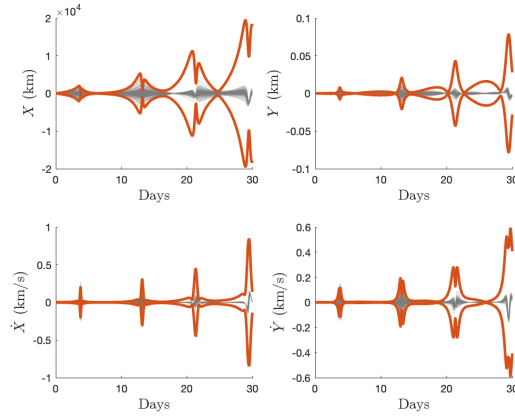
The third grid point tested was one with a high nonlinearity index. A grid point in the approaching zero velocity curve group was tested. The logarithm of the RMS of the nonlinearity index up to 30 days of the trajectory is 3.05. Figure (11) shows the distance from the mean of the 5000 sphere points as well as the 3σ bounds based on the covariance.



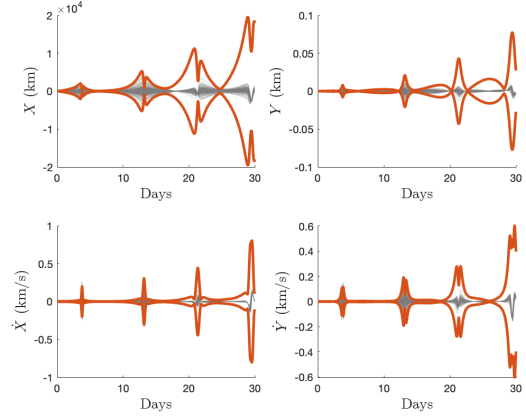
(a) Linear Propagation



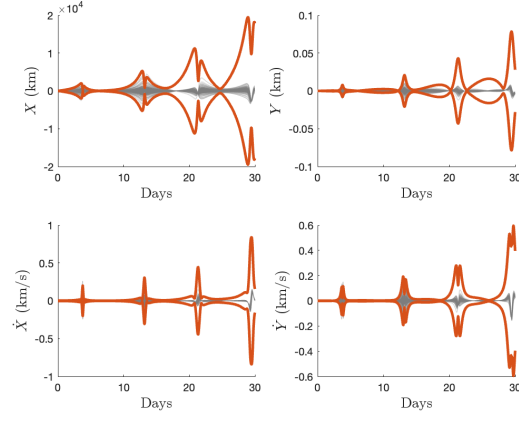
(b) Unscented Transform



(c) CUT 4

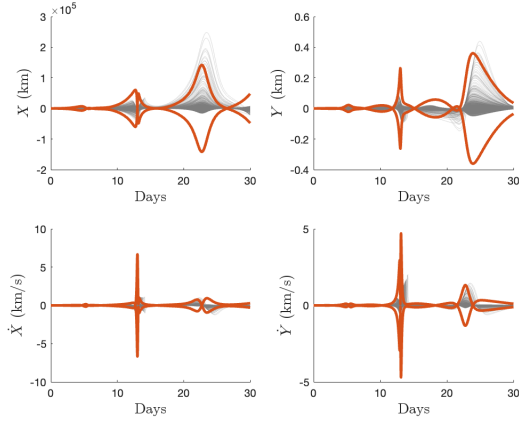


(d) CUT 6

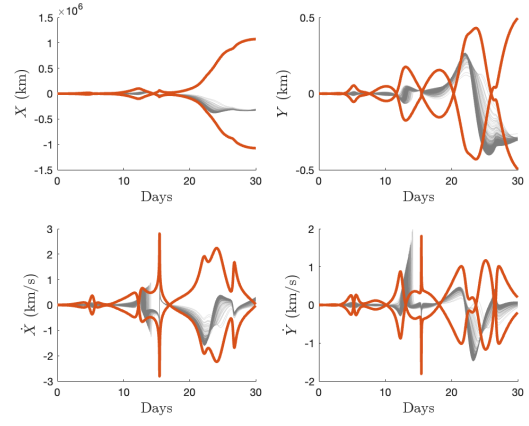


(e) CUT 8

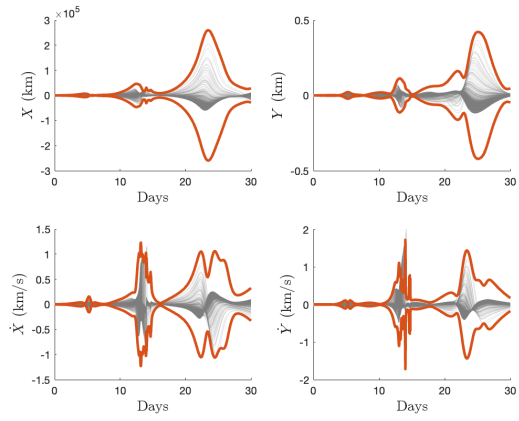
Figure 9: Uncertainty propagation performance for grid point 1739 (low nonlinearity)



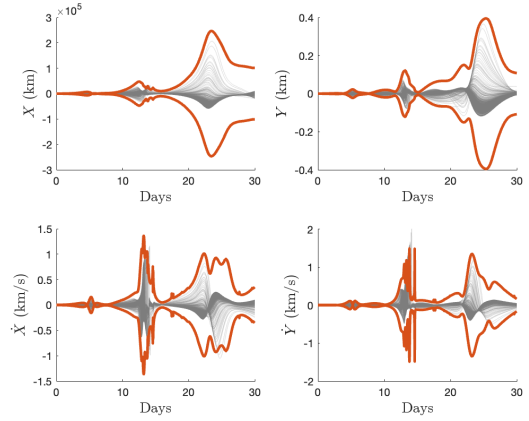
(a) Linear Propagation



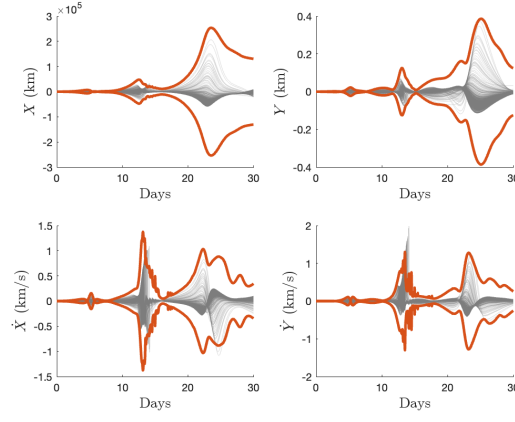
(b) Unscented Transform



(c) CUT 4

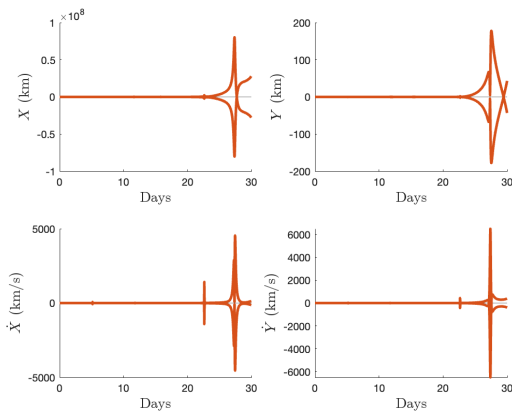


(d) CUT 6

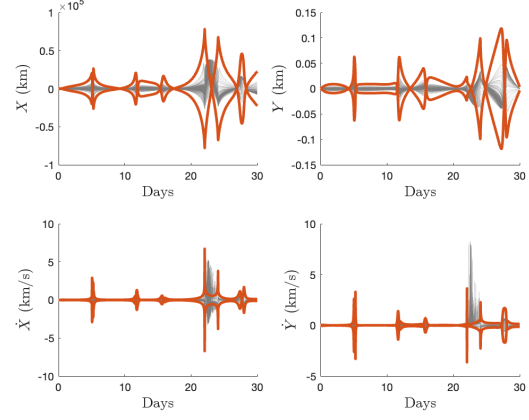


(e) CUT 8

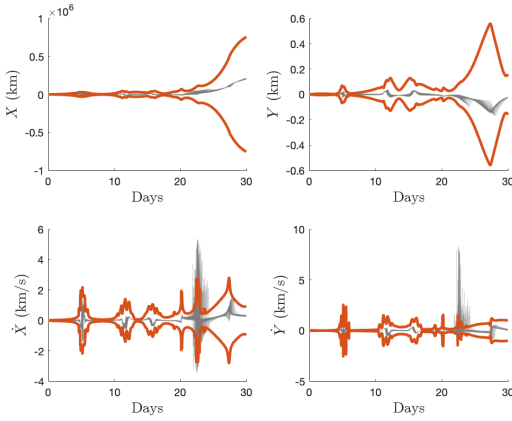
Figure 10: Uncertainty propagation performance for grid point 308 (medium nonlinearity)



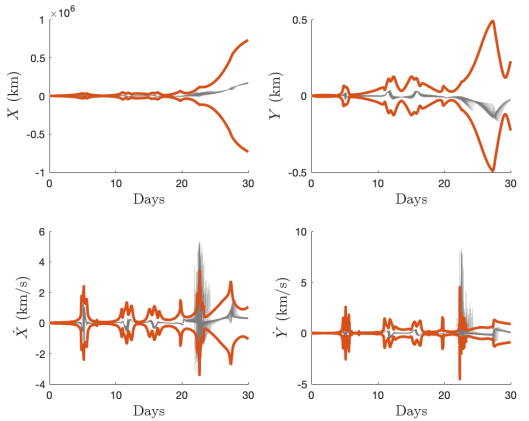
(a) Linear Propagation



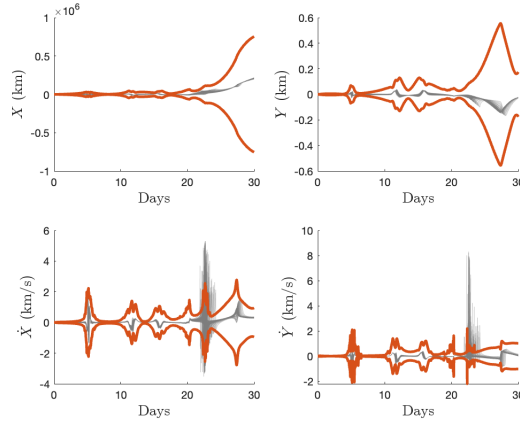
(b) Unscented Transform



(c) CUT 4



(d) CUT 6



(e) CUT 8

Figure 11: Uncertainty propagation performance for grid point 858 (high nonlinearity)

Across the grid points tested, there are a few notable trends:

- The linear propagation covariance becomes far too large. For the low and high nonlinearity cases, the range was so large the 5000 perturbed points were a line. Oddly, for the medium nonlinearity case, its performance was far better.
- The unscented transform method performed well. UT general estimated a small covariance. For the high nonlinearity case, UT performed very well.
- Compared to the UT, CUT was a conservative. CUT performed very well for low and medium nonlinearity. For high nonlinearity, the provided range was larger than needed, but reasonable.
- The differences between the CUT 4, 6, and 8 were small. It is difficult to tell the difference visually. CUT 4, with its low computational time, performs well.

SUMMARY

The study explores the growth of state errors and the consistency of uncertainty propagation algorithms in cislunar space. It quantifies state error growth in the Circular Restricted Three-Body Problem (CR3BP) and evaluates the performance of different algorithms. The Nonlinearity Index and the Finite Time Lyapunov Exponent are used as error propagation metrics. The results show that nonlinearity in the cislunar region leads to significant trajectory deviations, especially in close proximity to the Earth or Moon.

Five uncertainty propagation algorithms were tested: Linear Propagation, Unscented Transform (UT), and Conjugate Unscented Transform (CUT) with several variants (CUT 4, 6, and 8). Linear Propagation is simplistic and inconsistent. UT performs well across the different cases. CUT methods demonstrate robust performance, particularly in regions of low and medium nonlinearity. CUT was conservative for the high nonlinearity case. The computational demands do increase with higher-order variants. The study concludes that UT and CUT methods outperform linear propagation in capturing uncertainties in nonlinear environments, providing valuable insights for mission design, long-term surveillance, and spacecraft navigation in cislunar space. To better distinguish between UT and CUT, more analysis is needed.

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